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**Final Exam**  
**International Campus**  
**DDS Program**  
**Biochemistry 2**  
**February 2025 semester**

**Name and Surname:**

**Student ID Number:**

**Course Coordinator: Dr. Panahi**

**Duration of Exam: 64 Minutes**

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| MCQ Question | 64 |
|--------------|----|

**Directions:**

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil

1) Which one of the following is a characteristic of lingual lipase?

- a. Activate at an alkaline pH
- ✓ b. Activate at an acidic pH
- c. Hydrolyze long-chain fatty acids from TAG
- d. Activate in the mouth

2) All of the following are the roles of cholecystokinin (CCK), EXCEPT:

- a. release of bile salts
- b. release of pancreatic lipases
- c. decrease of gastric motility
- ✓ d. release of bicarbonate from the pancreas

3) All of the following indicate the contents of mixed micelles, EXCEPT:

- a. fat-soluble vitamins
- b. bile salts
- c. triacylglycerol
- ✓ d. free fatty acids ✓

4) In cystic fibrosis patients, all of the following are major complications, EXCEPT:

- ✓ a. pancreatic insufficiency ✓
- b. chronic lung infections
- c. male infertility
- d. cholestasis

5) ..... activate fatty acids for participation in the metabolic process.

- a. CoA ✓
- b. NADH
- c. FADH<sub>2</sub>
- d. NADPH

6) Which of the following pathways is correctly matched with its site in the cell?

- a. fatty acid synthesis, mitochondria

- b. ketone body synthesis, smooth endoplasmic reticulum (SER) ✓
- c.  $\beta$ -oxidation, mitochondria
- d. cholesterol synthesis, smooth endoplasmic reticulum (SER)

7) Ketone bodies ....

- a. are the main fuel source
- b. are used by the brain
- c. are important in the fed state
- ✓ d. are decreased in uncontrolled type 1 diabetes mellitus

8) Which of the following will NOT be produced in  $\beta$ -oxidation?

- a. NADH
- b. FADH<sub>2</sub>
- ✓ c. acetyl-CoA
- d. NADPH

9) If the cholesterol influx in the liver is higher than the cholesterol efflux, which of the following will happen?

- a. fatty liver disease
- b. pancreatic insufficiency
- c. plaque formation
- d. cholestasis ✓

10) The cholesterol is metabolized in the human body by conversion to.....

- a. H<sub>2</sub>O
- b. bile acids
- c. CO<sub>2</sub>
- d. acetyl-CoA ✓

11) All of the following are HDL functions, EXCEPT:

- a. apolipoprotein supply
- b. non-esterified cholesterol uptake

- ☒ c. cholesterol transformation from liver to peripheral tissues
- d. reverse cholesterol transport

12) Which of the following is observed in **nascent VLDL**?

- a. Apo B-100
- b. Apo C-II
- c. Apo E
- ☒ d. Apo B-48

13. Which of the following statements about transamination reactions is correct?

- a) Transamination reactions involve ATP hydrolysis.
- b) Transamination reactions are irreversible.
- c) Transamination reactions require  $\text{NAD}^+$  or  $\text{NADP}^+$ .
- ☒ d) Transamination reactions require pyridoxal-5'-phosphate (PLP)

14. Which amino acid are exclusively ketogenic?

- ☒ a) Alanine and cysteine
- b) Histidine and valine
- c) Leucine and lysine
- d) Tyrosine and proline

15. Maple syrup disease is a disorder due to a deficiency of pathway that degrades:

- a) Tyrosine
- ☒ b) Methionine
- c) Tryptophan
- d) Isoleucine

16. Identify the amino acid that participates in methyl group transfer:

- a) Methionine
- b) Lysine
- c) Serine
- ☒ d) Valine

17. Histamine, a chemical mediator of allergies and anaphylaxis, is synthesized from amino acid Histidine by which of the following reaction?

- a) Demethylation
- ☒ b) Hydrogenation
- c) Decarboxylation
- d) Oxygenation

18. Amino acids are released into the portal system by:

- a) Pinocytosis
- b) Active transport ✓
- c) Facilitated diffusion
- d) Passive transport

19. For the ATP-dependent proteasome system, which of the following proteins is required to degrade proteins?

- a) Cyclin
- b) Cytochrome
- c) Ubiquitin
- ✓d) Histone

20. Which of the following enzyme is deficient in alkaptonuria?

- a) Homogentisate dioxygenase
- ✓b) p-Hydroxyphenyl- pyruvate dioxygenase
- c) Phenylalanine hydroxylase
- d) Tyrosinase

21. Which of the following enzyme is small enough to pass through the glomeruli of the kidneys?

- a) Acid Phosphatase
- b) Amylase
- c) Nucleotidase
- ✓d) Aminotransferase

22. What enzyme in the body is measured to diagnose rhabdomyolysis?

- ✓a) Lipase
- b) Nucleotidase
- c) Creatine kinase
- d) Glutaryltransferase

23. Slight or moderate increases in serum acid phosphatase activity were found in all the following diseases, except:

- ✓a) Paget disease
- b) Hyperparathyroidism with skeletal involvement
- c) Nonalcoholic Fatty Liver Disease ✓
- d) Malignant invasion of the bones

24. All of the following are pancreatic specific enzyme, except?

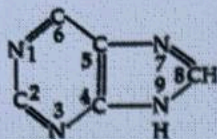
- a) Lipase
- ✓b) Aldolase
- c) Trypsin
- d) Amylase

25. Which of the following enzymes is required for the regeneration of reduced glutathione?

- ☒ a. Glucose 6 phosphatase
  - b. Phosphoglucomutase
  - c. Lactate dehydrogenase
  - d. Glucose 6 phosphate dehydrogenase
26. Pentose phosphate pathway is responsible to make all the following EXCEPT .....?
- a. Energy
  - b. Fructose
  - c. Ribose
  - ☒ d. NADPH
27. In Glycolysis, the synthesis of ATP is catalyzed by.....
- a. Phosphohexose isomerase
  - ☒ b. Glucokinase
  - c. Pyruvate kinase
  - d. Aldolase
28. Which of the following is a high energy molecule?
- a. dihydroxyacetone-P
  - b. glyceraldehydes 3- P
  - c. 1,3-biphosphoglycerate
  - ☒ d. 2 phosphoglycerate
29. Deficiency of which of the following molecules stops glycolysis?
- a. FAD+
  - b. FADH
  - ☒ c. NAD+
  - d. NADH
30. Which of the following enzymes is important in glycogenolysis?
- ☒ a. Phosphorylase
  - b. Hexokinas
  - c. Glucokinase
  - d. Phosphatase
31. Which glucose transporter is active in liver tissues?
- ☒ a. GLUT-1
  - b. GLUT-2
  - c. GLUT-3
  - d. GLUT-4
32. All the following enzymes are important in gluconeogenesis except:
- a. PEPCK
  - b. Fructose 1,6 biphosphatase

- c. Glucose 6 phosphatase ✓
  - d. Phosphorylase
- 33. How many ATP molecules are required for gluconeogenesis?
  - ✓ a. 1
  - b. 2
  - c. 3
  - d. 4
- 34. Where does TCA cycle take place?
  - a. Cytosol
  - b. Mitochondria
  - ✓ c. Nucleus
  - d. Cell membrane
- 35. Oxidation of one glucose molecule yields how many ATP molecules?
  - a. 26-28
  - ✓ b. 30-32
  - c. 34-36
  - d. 38-40
- 36. The end product of glycolysis under AEROBIC conditions is.....?
  - ✓ a. Fructose-6-phosphate
  - ✓ b. Pyruvic acid
  - c. Oxaloacetic acid
  - d. Lactic acid
- 37. .... donates its phosphate to ADP to yield ATP during muscle exertion
  - a. Creatine phosphate
  - ✓ b. Creatine kinase
  - c. Guanidoacetate
  - d. Arginine glycine amidinotransferase
- 38. Jaundice is caused by all of the following, except for...?
  - ✓ a. High amount of fatty acid
  - b. Impaired conjugation of bilirubin
  - c. obstruction of biliary tree
  - d. Excess RBC lysis
- 39. In Lesch-Nyhan syndrome
  - a. There is a defect or lack in HGPRT enzyme
  - b. There is an excess in the HGPRT enzyme
  - ✓ c. HGPRT enzyme is not involved in the disease
  - d. None of the above

40. According to the biosynthetic origin of purine ring, nitrogen atom N1 originates from?



- ☒ a. Glutamine
  - ☐ b. Glycine
  - ☐ c. Aspartic acid
  - ☐ d. Formate
41. Which of the following is necessary to convert heme into bilirubin in the body?
- ☐ A. Glucuronic acid and Aminolevulinic acid
  - ☐ B. Carbon monoxide and glutamine
  - ☒ C. NADPH and oxygen
  - ☐ D. glucuronic acid and NADH
42. Inhibition of which of the following enzymes is used in the treatment of gout?
- ☐ A. Thymidylate synthetase
  - ☐ B. Adenosine deaminase
  - ☐ C. Xanthine oxidase
  - ☒ D. Ribonucleotide reductase
43. which step of the following is the regulatory step in the Heme synthesis done?
- ☐ A. Delta aminolevulinic formation
  - ☒ B. Conversion of delta-aminolevulinic acid to porphobilinogen
  - ☐ C. Cyclization of tetrapyrrole
  - ☐ D. Decarboxylation of uroporphyrinogen
44. 5- Fluorouracil inhibits which of the following enzymes?
- ☒ A. Ribonucleotide reductase
  - ☐ B. Dihydrofolate reductase
  - ☐ C. Thymidine synthase
  - ☐ D. Xanthine oxidase
45. Which of the complexes of the respiratory chain does not act as a proton pump?
- ☐ A. Succinate Q Reductase
  - ☒ B. NADH dehydrogenase
  - ☐ C. Cytochrome C oxidase
  - ☐ D. Cyt C reductase
46. The ATP/ADP transporter in the mitochondrial inner membrane is inhibited by .....
- ☒ A. Oligomycin
  - ☐ B. CN-
  - ☐ C. Atracetylloside

- D. Barbiturates
47. Which of the transporters of the respiratory chain has the lowest oxidizing power?
- ✓ A. FMN
  - B. Coenzyme Q
  - C. Cytb
  - D. Cyt c
48. In the respiratory chain, cytochrome a is located in complex.....
- A. I
  - ✓ B. II
  - C. III
  - D. IV
49. In the feeding state, which of the following is correct?
- ✓ A. Decreased fatty acid synthase activity
  - B. Decreased acetyl-CoA carboxylase activity
  - C. Decreased carnitine palmitoyltransferase I activity
  - D. Decreased HMG-CoA reductase activity
50. Which enzyme is responsible for the hydrolysis of stored triacylglycerols in adipose tissue?
- A. Acetyl coenzyme A carboxylase (ACC)
  - B. Fatty acid synthase (FAS)
  - ✓ C. Hormone-sensitive lipase (HSL)
  - D. Lipoprotein lipase (LPL)
51. What is the primary role of the liver during fasting?
- A. Glycogen synthesis
  - B. Gluconeogenesis
  - C. Fatty acid oxidation
  - ✓ D. Ketone body synthesis
52. Which transporter is responsible for glucose uptake in muscle and adipose tissue?
- ✓ A. GLUT-1
  - B. GLUT-2
  - C. GLUT-3
  - D. GLUT-4
53. Which type of hormone binds intracellular receptors and directly regulates gene transcription?
- ✓ a) Steroids
  - b) Catecholamines

- c) Peptides
- d) Glycoproteins

54. Nephrogenic diabetes insipidus is caused by:

- ☒ a) Lack of ADH secretion
- b) Defect in V2 receptor signaling
- c) Increased insulin sensitivity
- d) Overexpression of phospholipase C

55. Sex hormone-binding globulin (SHBG) transports which hormones?

- ☒ a) Thyroid hormones
- b) Testosterone and estrogen
- c) Cortisol and aldosterone
- d) Insulin and glucagon

56. Cholesterol serves as a precursor for:

- a) Cortisol
- ☒ b) Insulin
- c) Epinephrine
- d) Glucagon

57. Which of the following symptoms usually appear in people with type 1 diabetes?

- a) Gaining weight quickly, and weakness
- b) Frequent urination, and excessive thirst
- c) decreased hunger and blurred vision
- ☒ d) High blood pressure

58. What triggers the secretion of glucagon from pancreatic alpha cells?

- a) Hyperglycemia
- ☒ b) Hypoglycemia
- c) High insulin levels
- d) High GLP-1 levels

59. In type 2 diabetes, the initial pathophysiological abnormality is most commonly:

- ☒ a) Beta-cell destruction
- b) Autoimmune attack
- c) Insulin resistance
- d) Increased glucagon clearance

60. Inositol 1,4,5-trisphosphate and Diacylglycerol are second messengers generated by the activation of which enzyme?

- a) Adenyl cyclase
- b) Phospholipase C
- ✓ c) Guanylyl cyclase
- d) Tyrosine kinase

61. A defect in 5-alpha reductase leads to which of the following?

- ✓ a) Impaired conversion of testosterone to dihydrotestosterone
- b) Increased cortisol synthesis
- c) Increased glucagon secretion
- d) Excessive insulin signaling

62. Which type of hormone binds intracellular receptors?

- a) Steroids
- b) Catecholamines
- c) Peptides
- ✓ d) Glycoproteins

63. Which of the following hormones acts through cAMP as a second messenger?

- ✓ a) Insulin
- b) Glucagon
- c) Growth hormone
- d) Cortisol

64. Which hormone is NOT stored in granules but bound to thyroglobulin?

- a) Epinephrine
- b) Thyroid hormones
- ✓ c) Insulin
- d) Growth hormone